ACTA | AI LAB

UNIVERSITEIT VAN AMSTERDAM
Data Science Centre

A PHD COLLABORATION, YEAR THREE

AI for Dentistry

From research to clinical translation

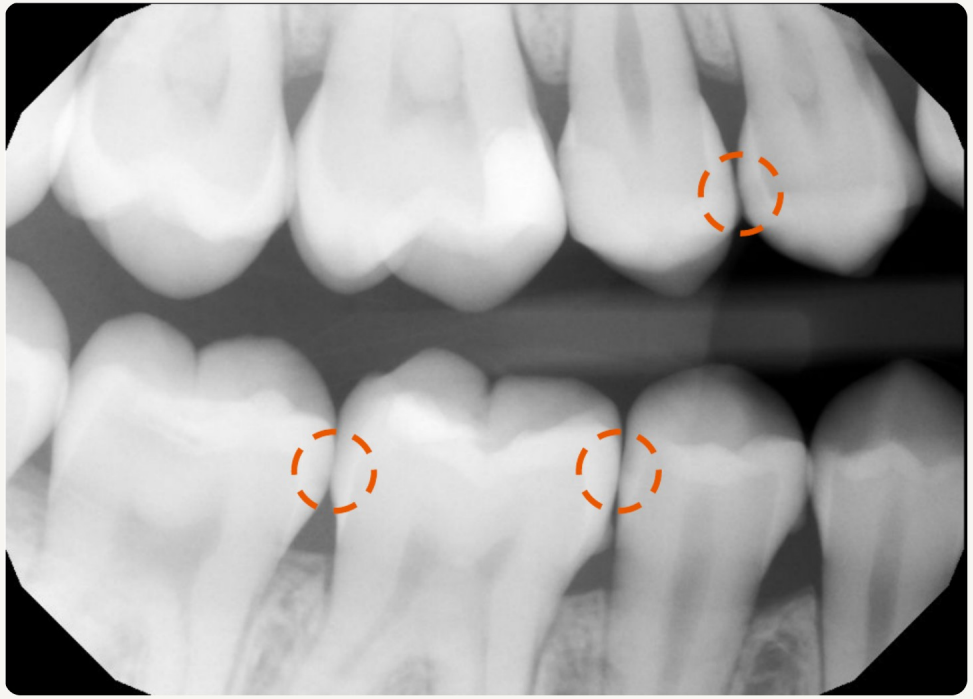
HAVA Lab | ACTA AI Lab × UvA Data Science Centre

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THE CLINICAL PROBLEM

Early caries are easy to miss



Circles mark proximal lesion sites · ACTA bitewing dataset

- Proximal lesions sit between teeth, so they are hard to see on a 2D bitewing.
- Initial-stage lesions are the hardest to diagnose.
- AI can support dentists, if it learns from enough varied, well-annotated data.

How do we make AI for caries lesion detection ready for the clinic?

ROADMAP

Bottlenecks between AI research and the clinic

1

Data

Too varied, too little

OUR WORK

Laboratory scans as extra training data

Synthetic bitewings, with annotations

2

Robustness

Models break when images change

OUR WORK

A hyperbolic U-Net (MIDL 2026)

A lighter, faster version for the clinic

3

Trust

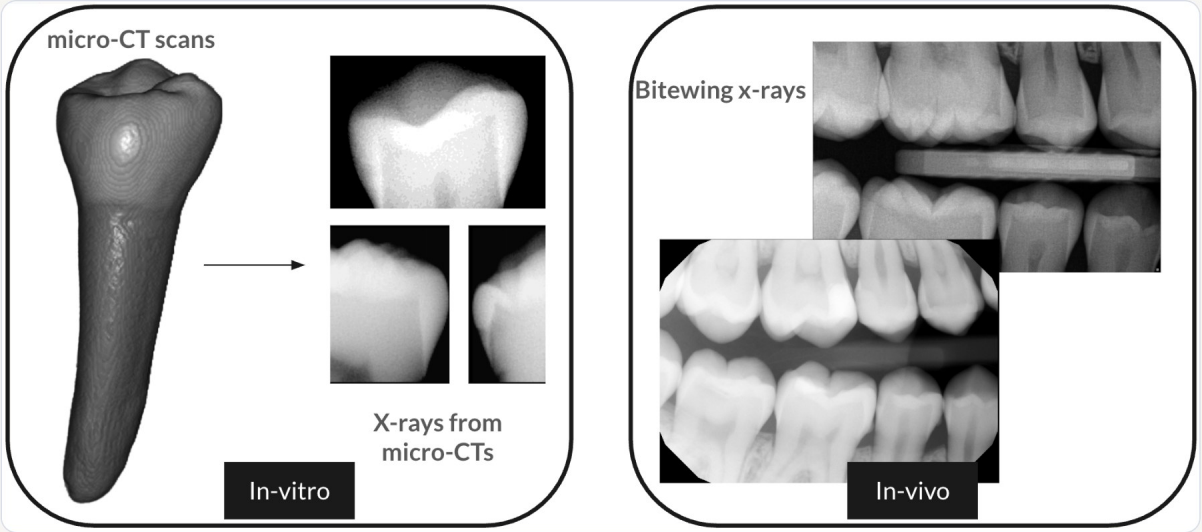
Accuracy is not patient benefit

OUR WORK

A patient-centred impact score for comparing models

ACT 1 · DATA

Can micro-CT scans help AI read bitewings?



Laboratory (in-vitro): X-rays from micro-CT. Clinical (in-vivo): bitewings.

THREE WAYS TO USE LABORATORY DATA

Pre-training
Learn from laboratory images first

Knowledge distillation
A laboratory-trained model guides clinical training

Both combined

Tested on 369 clinical proximal regions · nnU-Net and YOLO · 5 folds × 3 random seeds

ACT 1 · DATA

Adding micro-CT data did not beat bitewings

nnU-Net trained with	Sensitivity	Specificity	Dice-Positive
Clinical bitewings only	0.536	0.896	0.276
Micro-CT pre-training, then clinical bitewings	0.534	0.849	0.260
Micro-CT AI model guides clinical bitewings	0.520	0.910	–

Means over three seeds; test set of 369 clinical proximal regions. YOLO: laboratory pre-training raised sensitivity (0.195 vs 0.165) at lower specificity.

Pre-training lowered accuracy

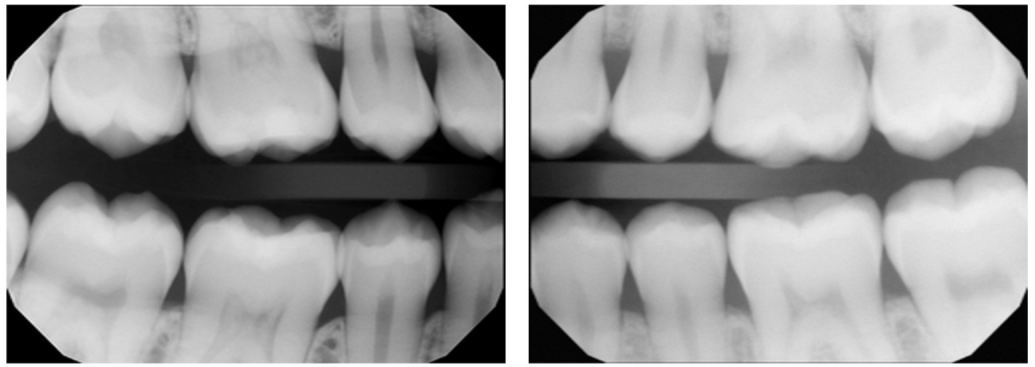
nnU-Net regional accuracy: OR 0.65, 95% CI 0.50–0.85 (nominal P = 0.0018)

Why it still matters

A rigorous negative result. It points us toward better clinical-like data.

ACT 1 · DATA

Can you tell which is real?



Real/AI-generated bitewings

9 dentists took part in the pilot

They could not tell real from AI-generated

Rated acceptable for training AI and for dental education

Next: a large survey is launching

ACT 1 · DATA

Next: bitewings from lesion delineations

Lesion mask
location and stage

→

Generative model
conditioned on the mask

→

Synthetic bitewing
with its paired annotation

Train AI models
More robust diagnostic models from scarce data

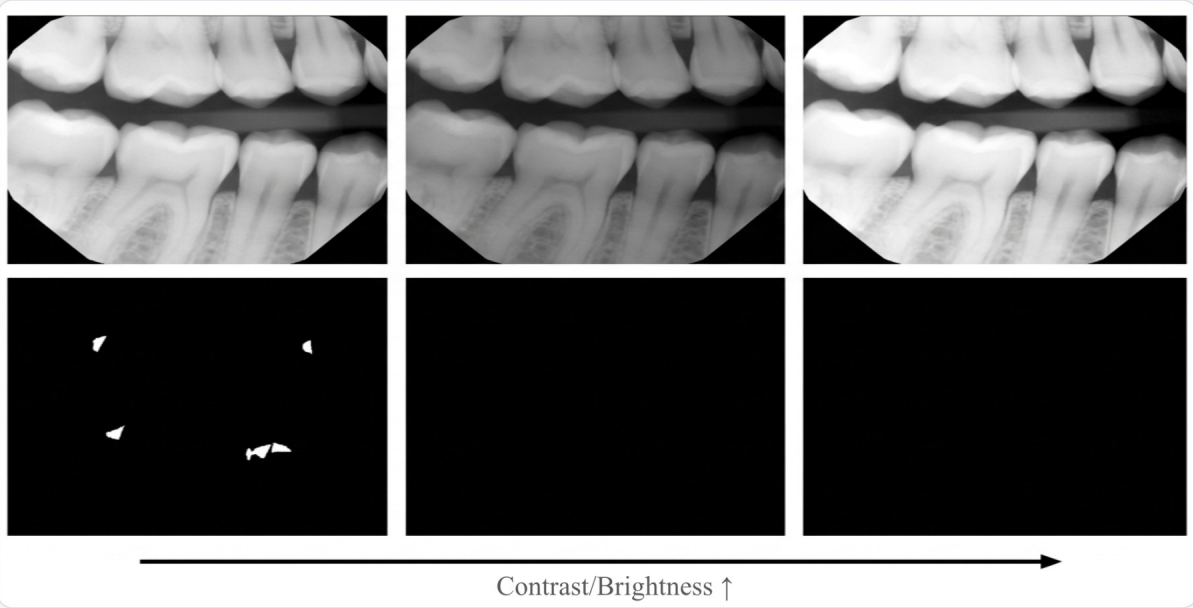
Lighter annotation load
Less manual labelling by dentists

Education
More cases for students to practise diagnosis

New collaboration, starting now

ACT 2 · ROBUSTNESS

Standard AI models break when images change

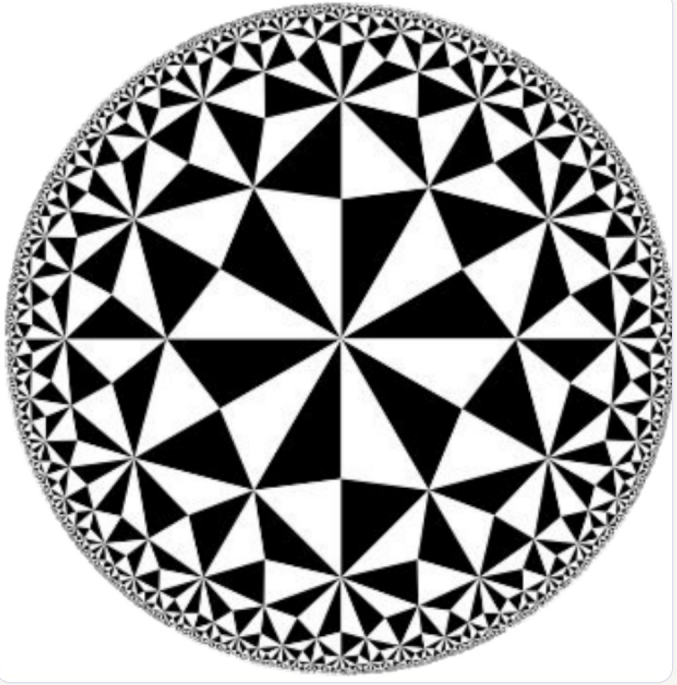


Proximal caries lesion segmentation

How do we make AI models robust to noise, contrast and scanner shifts?

ACT 2 · ROBUSTNESS

A new geometry for robustness



Poincaré disk: a model of hyperbolic space

The U-Net works in flat, Euclidean space. Hyperbolic space is curved, and models built in it can be more compact and have shown better robustness.

The first fully hyperbolic U-Net

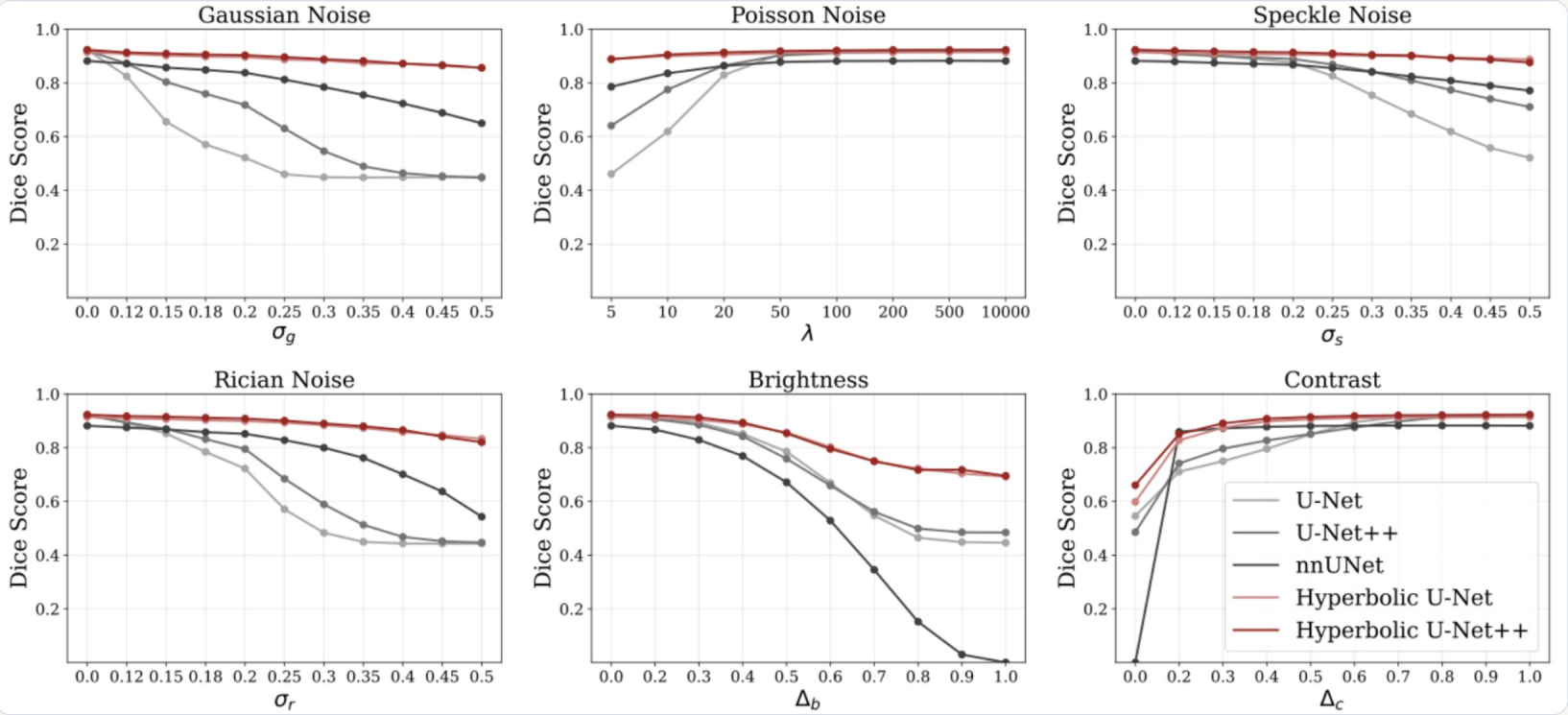
Hyperbolic transposed convolution

Hyperbolic bilinear upsampling

Newton approximation layer initialisation

Accepted at MIDL 2026

Hyperbolic U-Nets stay accurate under shifts



Red: hyperbolic. **Grey:** U-Net, U-Net++ and nnU-Net.

Dice score stays high as noise and brightness shifts grow.

ACT 2 · ROBUSTNESS

Now: a lighter, faster model for clinical settings

Done

Robust

Hyperbolic U-Net stays accurate under noise and brightness shifts.

In progress

Lighter and faster

Smaller model and shorter inference time, without losing robustness.

Goal

Clinical deployment

Run on real bitewings in real clinical settings.

We re-built our **Hyperbolic U-Net** model to be **6.5 x faster** and **6.0 x lighter**.

ACT 3 · TRUST

Accuracy is not patient benefit

Not every error costs the patient the same.

Our score weights each outcome by its consequence, in the context of how common the disease is.

We use it to compare training strategies, including the micro-CT-data study.

Work in progress: manuscript in development

	AI says lesion	AI says no lesion
Lesion present	**Early treatment** (benefit)	**Missed progression** (harm)
No lesion	**Unneccessary filling** (harm)	**Reassurance** (benefit)

THE COLLABORATION

What we built together

Universiteit van Amsterdam
Data Science Centre

Deep learning, geometry and generative models

ACTA | AI LAB

Clinical questions, annotations and dentists' judgement

Project	Status
Hyperbolic U-Net, robust segmentation	Accepted at MIDL 2026
Micro-CT data for caries lesion detection	Manuscript in preparation
Synthetic bitewings, 9-dentist pilot	Large survey launching
Generation from lesion masks	New collaboration starting
Lighter, faster hyperbolic U-Net	In progress
Patient-centred impact score	Manuscript in development

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Thank you

Questions, ideas and opinions are welcome

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